Traffic Flow Breakdown Prediction for the M50 Motorway in Ireland

Jack Geraghty[1], Paul Hurlet[2], Maha Riad[1], Zakaria Sabir[1], Robert Corbally[1], Fatemeh Golpayegani[1]
University College Dublin, Dublin, Ireland[1]
ENTPE National School of Public Works, Lyon France[2]

Introduction

- The M50 is Irelands busiest road.
- 145,000 vehicles on average per day.
- Currently operating Variable Speed Limits (VSL) and Lane Control Systems (LCS).
- Manually operated from control station.
- Move to automatic control requires real-time information:
  - Road conditions
  - Is there an imminent breakdown in traffic flow?
- Can this breakdown in traffic flow be predicted?

M50 Data

- Double-inductive loop spaced approximately 500m apart.
- Gather data such as:
  - Speed
  - Flow
  - Density
  - Occupancy
  - Vehicle length
- Resolution of 20s
- Weather sensors:
  - Road surface temperature
  - Visibility.
  - Precipitation

A Genetic Algorithm for Optimal Model Design

- Used an Artificial Neural Network (ANN)
- Formulated feature selection as an optimisation problem.
- Solved using a genetic algorithm
  - Optimised according to F1-Score
- Varied the number of input features
  - Number of upstream/downstream induction loops.
  - Which features to include for example Speed, Density, Day type LOS.
- 500 random ANNs per generation.
- Top 10 used in the next generation.
- Observations:
  - Hidden size had small impact on metrics.
  - Speed, density and speed variation appeared most often (>80%)

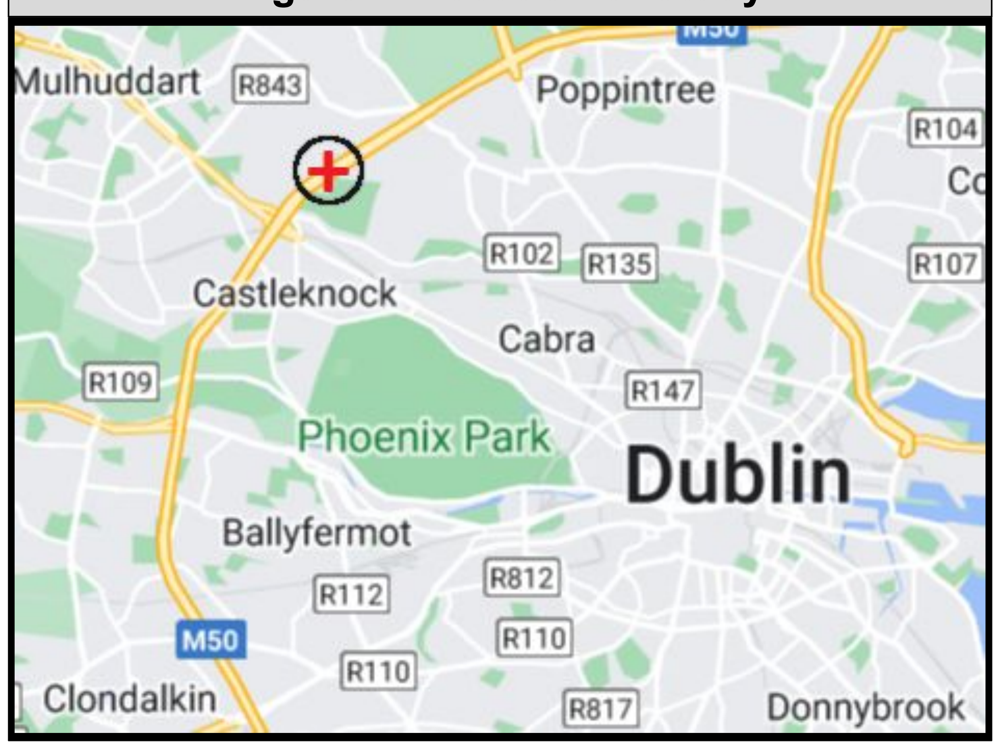
Table 1: Possible Input Features

Name	Values
Volume	Number of vehicles >=0
Occupancy	Between 0 and 100 [%/]
Speed	Between 0 and 150 [km/h]
Speed Variation	Between -150 and 150 [km/h]
Vehicle Length	>=0 [m]
Flow	>=0 [veh/h]
Density	>=0 [veh/km]
Congested State	1 or 0 (True or False) **Density** > 22 or **Speed** <= 60
Level of Services	1 to 6 (See Table 2)

Experiments

- 20s data aggregated into 5-minute buckets.
- 105,120 data points.
- **Congested State:**
  - Greater than 22 vehicles per kilometre per lane.
  - Average speed less than 60Km/h.
- To prevent false positives:
  - Must last for two data points (10 minutes).
  - Must be at least 10 minutes apart from another occurrence of congestion.

Figure 1: Location of Study



- Focused on single loop
- Loop 5320 - Southbound
  - Significant historical data.
  - Experiences congestion almost daily.

Table 2: Level of Service (LOS) Bands

LOS	Density Range (vehicles/lane/km)	Description
A	0-7	Free Flow
B	7-11	Reasonable Free Flow
C	11-16	Stable Flow
D	16-22	Approaching Unstable Flow
E	22-28	Unstable Flow
F	28+	Force breakdown of flow

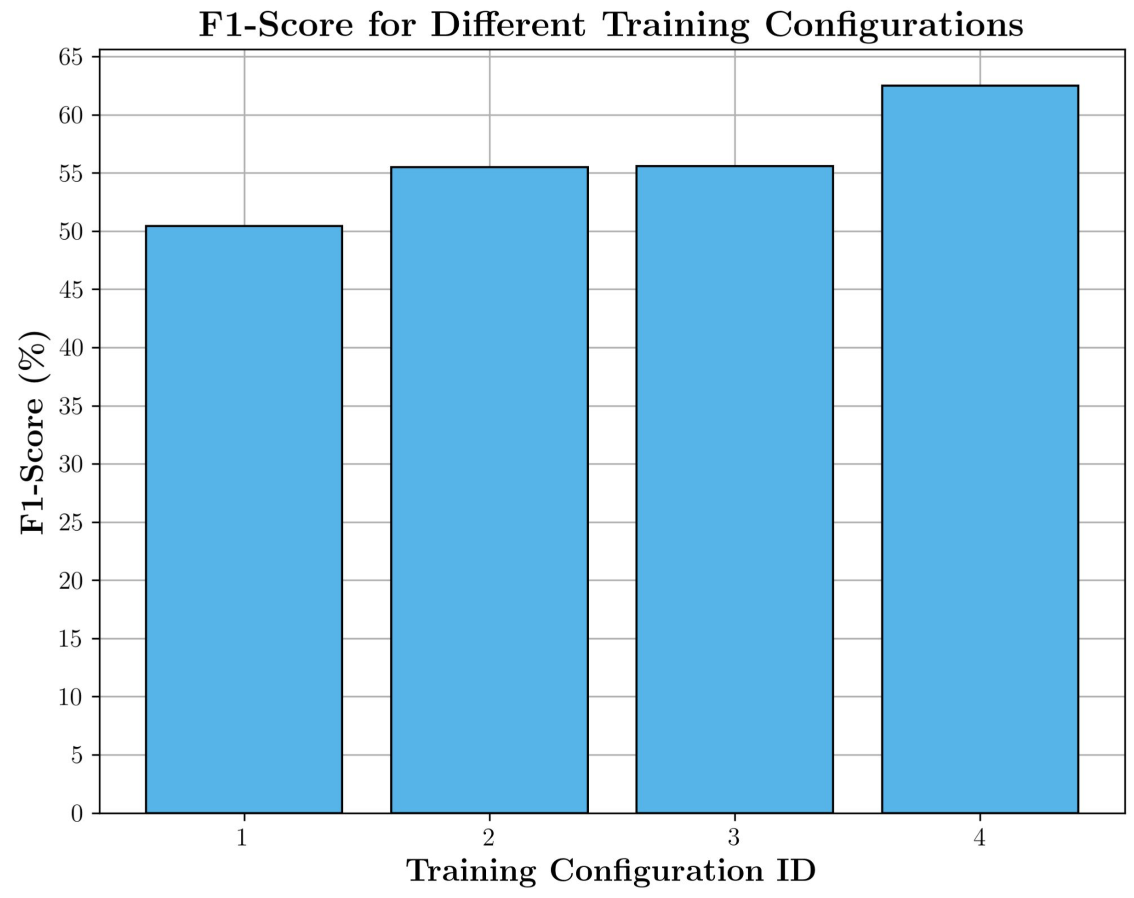
- Significant class imbalance: 95:5 towards uncongested samples.
- Tested variations on subsets of the full data set.
- Varying impact on performance.

Table 3: Training Configurations

ID	Description
1	All year
2	2hrs before congestion
3	Density > 21 and SC5S
4	2hrs before congestion and density > 21 and SCS5

SCS5: The first 5 minutes of congestion where the change from a smooth to a congested state occurs.

Results



- **Configuration 1:** Class imbalance significantly impacted learning.
- **Configuration 2:** Impact of class imbalance is lessened.
- **Configuration 3:** More false positives. Likely due to absence of samples at the density threshold (22).
- **Configuration 4:** Partially mitigates the short-comings of configurations 2 and 3.

Observations

- Increase in false positives in the 30 minutes prior to congestion.
- Increase in false negatives when a short period of congestion is closely followed by another.

Conclusion

- Most important factors:
  - Definition of congestion.
  - Data set design and wrangling.
- Model architecture does not significantly impact performance.
- Most available features do no significantly impact performance.
- Future works:
  - Reducing false positives before congestion.
  - Reducing the number of false negatives

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Contact Information

Jack Geraghty

Address: A0.12, School of Computer Science, UCD, Dublin, Ireland. D04V1W8

Tel: +353 876191424
Email: jack.geraghty@ucdconnect.ie

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